INSTALLING JAVA IN COLAB ENVIRONMENT

```
# Install Java (required for Hadoop)

!apt-get update
!apt-get install -y openjdk-11-jdk-headless
!java -version
```

```
Hit:4 http://archive.ubuntu.com/ubuntu jammy InRelease
Get:5 https://r2u.stat.illinois.edu/ubuntu jammy InRelease [6,555 B]
Get:6 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:7 https://cloud.r-project.org/bin/linux/ubuntu jammy-cran40/ Packages [85.2 kB]
Get:8 https://cli.github.com/packages stable/main amd64 Packages [357 B]
Get:9 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu jammy InRelease [18.1 kB]
Get:10 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:11 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [6,652 kB]
Get:12 https://ppa.launchpadcontent.net/ubuntugis/ppa/ubuntu jammy InRelease [24.6 kB]
Get:13 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 Packages [62.6 kB]
Get:14 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [3,780 kB]
Get:15 https://r2u.stat.illinois.edu/ubuntu jammy/main all Packages [9,813 kB]
Get:16 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [1,301 kB]
Get:17 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [4,120 kB]
Get:18 https://r2u.stat.illinois.edu/ubuntu jammy/main amd64 Packages [2,922 kB]
Get:19 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu jammy/main amd64 Packages [39.2 kB]
Get:20 http://archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 Packages [70.9 kB]
Get:21 http://archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [6,877 kB]
Get:22 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1,613 kB]
Get:23 https://ppa.launchpadcontent.net/ubuntugis/ppa/ubuntu jammy/main amd64 Packages [75.3 kB]
Fetched 37.9 MB in 7s (5,462 kB/s)
Reading package lists... Done
W: Skipping acquire of configured file 'main/source/Sources' as repository 'https://r2u.stat.illinois.edu/ubuntu jammy InR
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  openjdk-11-jre-headless
Suggested packages:
  openjdk-11-demo openjdk-11-source libnss-mdns fonts-dejavu-extra
  fonts-ipafont-gothic fonts-ipafont-mincho fonts-wqy-microhei
  | fonts-wqy-zenhei fonts-indic
The following NEW packages will be installed:
  openjdk-11-jdk-headless openjdk-11-jre-headless
0 upgraded, 2 newly installed, 0 to remove and 114 not upgraded.
Need to get 116 MB of archives.
After this operation, 258 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 openjdk-11-jre-headless amd64 11.0.30+7-1ubuntu1~22.04 [42
Get:2 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 openjdk-11-jdk-headless amd64 11.0.30+7-1ubuntu1~22.04 [73
Fetched 116 MB in 2s (68.2 MB/s)
Selecting previously unselected package openjdk-11-jre-headless:amd64.
(Reading database ... 117540 files and directories currently installed.)
Preparing to unpack .../openjdk-11-jre-headless_11.0.30+7-1ubuntu1~22.04_amd64.deb ...
Unpacking openjdk-11-jre-headless:amd64 (11.0.30+7-1ubuntu1~22.04) ...
Selecting previously unselected package openjdk-11-jdk-headless:amd64.
Preparing to unpack .../openjdk-11-jdk-headless_11.0.30+7-1ubuntu1~22.04_amd64.deb ...
Unpacking openjdk-11-jdk-headless:amd64 (11.0.30+7-1ubuntu1~22.04) ...
Setting up openjdk-11-jre-headless:amd64 (11.0.30+7-1ubuntu1~22.04) ...
update-alternatives: using /usr/lib/jvm/java-11-openjdk-amd64/bin/jjs to provide /usr/bin/jjs (jjs) in auto mode
update-alternatives: using /usr/lib/jvm/java-11-openjdk-amd64/bin/rmid to provide /usr/bin/rmid (rmid) in auto mode
update-alternatives: using /usr/lib/jvm/java-11-openjdk-amd64/bin/pack200 to provide /usr/bin/pack200 (pack200) in auto mo
update-alternatives: using /usr/lib/jvm/java-11-openjdk-amd64/bin/unpack200 to provide /usr/bin/unpack200 (unpack200) in a
Setting up openjdk-11-jdk-headless:amd64 (11.0.30+7-1ubuntu1~22.04) ...
update-alternatives: using /usr/lib/jvm/java-11-openjdk-amd64/bin/rmic to provide /usr/bin/rmic (rmic) in auto mode
update-alternatives: using /usr/lib/jvm/java-11-openjdk-amd64/bin/jaotc to provide /usr/bin/jaotc (jaotc) in auto mode
openjdk version "17.0.17" 2025-10-21
OpenJDK Runtime Environment (build 17.0.17+10-Ubuntu-122.04)
OpenJDK 64-Bit Server VM (build 17.0.17+10-Ubuntu-122.04, mixed mode, sharing)
```

DOWNLOADING APACHE HADOOP ON COLAB

Disclimer : Pls check the version before downloading

```
# Download Hadoop

!wget https://downloads.apache.org/hadoop/common/hadoop-3.4.1/hadoop-3.4.1.tar.gz
```

```
--2026-03-06 13:10:18--  https://downloads.apache.org/hadoop/common/hadoop-3.4.1/hadoop-3.4.1.tar.gz
Resolving downloads.apache.org (downloads.apache.org)... 88.99.208.237, 135.181.214.104, 2a01:4f9:3a:2c57::2, ...
Connecting to downloads.apache.org (downloads.apache.org)|88.99.208.237|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 974002355 (929M) [application/x-gzip]
Saving to: ‘hadoop-3.4.1.tar.gz’

hadoop-3.4.1.tar.gz 100%[===================>] 928.88M  24.4MB/s    in 40s
```

```
2026-03-06 13:10:58 (23.2 MB/s) - ‘hadoop-3.4.1.tar.gz’ saved [974002355/974002355]
```

EXTRACTING HADOOP

```
# Extract Hadoop

!tar -xzvf hadoop-3.4.1.tar.gz
```

```
Streaming output truncated to the last 5000 lines.
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/cosn/auth/package-summary.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/cosn/auth/package-use.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/cosn/package-frame.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/cosn/package-summary.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/cosn/package-use.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/FSError.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/class-use/
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/class-use/Constants.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/class-use/RemoteFileChangedException.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/class-use/AWSServiceIOException.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/class-use/SimpleAWSCredentialsProvider.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/class-use/UnknownStoreException.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/class-use/CredentialInitializationException.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/class-use/AWSS3IOException.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/class-use/SharedInstanceCredentialProvider.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/class-use/NoVersionAttributeException.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/class-use/TemporaryAWSCredentialsProvider.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/class-use/S3A.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/auth/
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/auth/class-use/
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/auth/class-use/AssumedRoleCredentialProvider.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/auth/class-use/IAMInstanceCredentialsProvider.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/auth/delegation/
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/auth/delegation/class-use/
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/auth/delegation/class-use/DelegationConstants.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/impl/
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/impl/class-use/
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/impl/class-use/MultiObjectDeleteException.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/select/
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/select/class-use/
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/select/class-use/SelectConstants.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/audit/
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/audit/class-use/
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/audit/class-use/S3LogParser.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/commit/
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/commit/class-use/
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/commit/class-use/CommitConstants.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/commit/class-use/CommitterStatisticNames.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/commit/magic/
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/commit/magic/class-use/
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/s3a/commit/magic/class-use/MagicS3GuardCommitter.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/FilterFileSystem.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/package-summary.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/package-use.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/BulkDeleteSource.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/FileChecksum.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/FileAlreadyExistsException.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/GlobalStorageStatistics.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/audit/
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/audit/class-use/
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/audit/class-use/CommonAuditContext.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/audit/package-tree.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/audit/package-frame.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/audit/package-summary.html
hadoop-3.4.1/share/doc/hadoop/api/org/apache/hadoop/fs/audit/package-use.html
```

ENVIRONMENTAL SETUP

Similar to environmental variable setup in ur local machines

```
# Set up environment variables for Java and Hadoop

import os
os.environ['JAVA_HOME'] = '/usr/lib/jvm/java-11-openjdk-amd64'
os.environ['HADOOP_HOME'] = '/content/hadoop-3.4.1'
os.environ['PATH'] = os.environ['HADOOP_HOME'] + '/bin:' + os.environ['PATH']

!echo $JAVA_HOME
!echo $HADOOP_HOME
```

```
/usr/lib/jvm/java-11-openjdk-amd64
/content/hadoop-3.4.1
```

SIMPLE EXERCISE USING MOVIELENS DATASET

```
# Download MovieLens 100k dataset

!wget https://files.grouplens.org/datasets/movielens/ml-100k.zip
!unzip ml-100k.zip
```

```
--2026-03-06 13:11:35--  https://files.grouplens.org/datasets/movielens/ml-100k.zip
Resolving files.grouplens.org (files.grouplens.org)... 128.101.96.204
Connecting to files.grouplens.org (files.grouplens.org)|128.101.96.204|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 4924029 (4.7M) [application/zip]
Saving to: ‘ml-100k.zip’

ml-100k.zip         100%[===================>]   4.70M  14.6MB/s    in 0.3s

2026-03-06 13:11:36 (14.6 MB/s) - ‘ml-100k.zip’ saved [4924029/4924029]

Archive:  ml-100k.zip
   creating: ml-100k/
  inflating: ml-100k/allbut.pl
  inflating: ml-100k/mku.sh
  inflating: ml-100k/README
  inflating: ml-100k/u.data
  inflating: ml-100k/u.genre
  inflating: ml-100k/u.info
  inflating: ml-100k/u.item
  inflating: ml-100k/u.occupation
  inflating: ml-100k/u.user
  inflating: ml-100k/u1.base
  inflating: ml-100k/u1.test
  inflating: ml-100k/u2.base
  inflating: ml-100k/u2.test
  inflating: ml-100k/u3.base
  inflating: ml-100k/u3.test
  inflating: ml-100k/u4.base
  inflating: ml-100k/u4.test
  inflating: ml-100k/u5.base
  inflating: ml-100k/u5.test
  inflating: ml-100k/ua.base
  inflating: ml-100k/ua.test
  inflating: ml-100k/ub.base
  inflating: ml-100k/ub.test
```

```
# Install mrjob library for MapReduce

!pip install mrjob
```

```
Collecting mrjob
  Downloading mrjob-0.7.4-py2.py3-none-any.whl.metadata (7.3 kB)
Requirement already satisfied: PyYAML>=3.10 in /usr/local/lib/python3.12/dist-packages (from mrjob) (6.0.3)
Downloading mrjob-0.7.4-py2.py3-none-any.whl (439 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 439.6/439.6 kB 8.1 MB/s eta 0:00:00
Installing collected packages: mrjob
Successfully installed mrjob-0.7.4
```

Process Creation

```
# Creating process to use MRJOB using Python

# Must know the structure of the dataset:
# First column reference to userID
# Second column reference to movieID
# Third column reference to rating
# Fourth column reference to timestamp

!head /content/ml-100k/u.data -n 10
```

```
196	242	3	881250949
186	302	3	891717742
22	377	1	878887116
244	51	2	880606923
166	346	1	886397596
298	474	4	884182806
115	265	2	881171488
253	465	5	891628467
305	451	3	886324817
6	86	3	883603013
```

```
%%writefile RatingBreakdown.py
# import modules
from mrjob.job import MRJob
from mrjob.step import MRStep
```

```
# create class inherited from MRJob
class RatingBreakdown(MRJob):
  # assign steps, first mapper last reducer
  def steps(self):
    return [
            MRStep(mapper=self.mapper_get_rating,
                   reducer=self.reducer_count_ratings)
    ]

  # creating mapper, assigning attributes from dataset
  def mapper_get_rating(self, _, line):
    (userID, movieID, rating, timestamp) = line.split('\t')
    # assign like the key rating and assign each row value 1
    yield rating, 1

  # creating reduce, sum
  def reducer_count_ratings(self, key, values):
    # in function of each key we sum values
    yield key, sum(values)

# run script
if __name__ == '__main__':
  RatingBreakdown.run()
```

```
Writing RatingBreakdown.py
```

```
# Running the process

!python RatingBreakdown.py /content/ml-100k/u.data > results.txt
```

```
No configs found; falling back on auto-configuration
No configs specified for inline runner
Creating temp directory /tmp/RatingBreakdown.root.20260306.131143.517020
Running step 1 of 1...
job output is in /tmp/RatingBreakdown.root.20260306.131143.517020/output
Streaming final output from /tmp/RatingBreakdown.root.20260306.131143.517020/output...
Removing temp directory /tmp/RatingBreakdown.root.20260306.131143.517020...
```

Result

```
!cat results.txt
```

```
"5"	21201
"4"	34174
"1"	6110
"2"	11370
"3"	27145
```

HANDS-ON

EXERCISE 1 : WORD COUNT

```
!echo "big data hadoop" > data.txt
!echo "hadoop processes big data" >> data.txt
!echo "big data analytics" >> data.txt
```

```
!cat data.txt
```

```
big data hadoop
hadoop processes big data
big data analytics
```

```
!hdfs dfs -mkdir /exercise1
!hdfs dfs -put data.txt /exercise1
```

```
!hadoop jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-*.jar wordcount /exercise1 /output1
```

```
2026-03-06 13:19:18,857 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-03-06 13:19:19,232 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2026-03-06 13:19:19,232 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2026-03-06 13:19:19,762 INFO input.FileInputFormat: Total input files to process : 1
2026-03-06 13:19:19,812 INFO mapreduce.JobSubmitter: number of splits:1
2026-03-06 13:19:20,369 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local1061362476_0001
2026-03-06 13:19:20,369 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-03-06 13:19:20,631 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-03-06 13:19:20,633 INFO mapreduce.Job: Running job: job_local1061362476_0001
2026-03-06 13:19:20,652 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-03-06 13:19:20,663 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOut
```

```
2026-03-06 13:19:20,664 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-03-06 13:19:20,664 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output
2026-03-06 13:19:20,665 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapreduce.lib.output.FileOutputCo
2026-03-06 13:19:20,734 INFO mapred.LocalJobRunner: Waiting for map tasks
2026-03-06 13:19:20,735 INFO mapred.LocalJobRunner: Starting task: attempt_local1061362476_0001_m_000000_0
2026-03-06 13:19:20,787 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOut
2026-03-06 13:19:20,789 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-03-06 13:19:20,789 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output
2026-03-06 13:19:20,837 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2026-03-06 13:19:20,847 INFO mapred.MapTask: Processing split: file:/exercise1/data.txt:0+61
2026-03-06 13:19:21,183 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-03-06 13:19:21,183 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-03-06 13:19:21,183 INFO mapred.MapTask: soft limit at 83886080
2026-03-06 13:19:21,183 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-03-06 13:19:21,183 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2026-03-06 13:19:21,192 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-03-06 13:19:21,217 INFO mapred.LocalJobRunner:
2026-03-06 13:19:21,220 INFO mapred.MapTask: Starting flush of map output
2026-03-06 13:19:21,220 INFO mapred.MapTask: Spilling map output
2026-03-06 13:19:21,221 INFO mapred.MapTask: bufstart = 0; bufend = 101; bufvoid = 104857600
2026-03-06 13:19:21,221 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 26214360(104857440); length = 37/65536
2026-03-06 13:19:21,254 INFO mapred.MapTask: Finished spill 0
2026-03-06 13:19:21,313 INFO mapred.Task: Task:attempt_local1061362476_0001_m_000000_0 is done. And is in the process of c
2026-03-06 13:19:21,318 INFO mapred.LocalJobRunner: map
2026-03-06 13:19:21,318 INFO mapred.Task: Task 'attempt_local1061362476_0001_m_000000_0' done.
2026-03-06 13:19:21,328 INFO mapred.Task: Final Counters for attempt_local1061362476_0001_m_000000_0: Counters: 18
        File System Counters
                FILE: Number of bytes read=281834
                FILE: Number of bytes written=996330
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
        Map-Reduce Framework
                Map input records=3
                Map output records=10
                Map output bytes=101
                Map output materialized bytes=72
                Input split bytes=89
                Combine input records=10
                Combine output records=5
                Spilled Records=5
                Failed Shuffles=0
                Merged Map outputs=0
                GC time elapsed (ms)=36
                Total committed heap usage (bytes)=394264576
        File Input Format Counters
```

```
!hdfs dfs -cat /output1/part-r-00000
```

```
analytics       1
big     3
data    3
hadoop  2
processes       1
```

EXERCISE 2 : LINE COUNT

```
!echo "machine learning" > lines.txt
!echo "big data" >> lines.txt
!echo "hadoop ecosystem" >> lines.txt
!echo "data science" >> lines.txt
```

```
!hdfs dfs -mkdir /exercise2
!hdfs dfs -put lines.txt /exercise2
```

```
!hadoop jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-*.jar wordcount /exercise2 /output2
```

```
2026-03-06 13:28:27,097 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-03-06 13:28:27,320 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2026-03-06 13:28:27,320 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2026-03-06 13:28:27,759 INFO input.FileInputFormat: Total input files to process : 1
2026-03-06 13:28:27,804 INFO mapreduce.JobSubmitter: number of splits:1
2026-03-06 13:28:28,200 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local487643619_0001
2026-03-06 13:28:28,200 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-03-06 13:28:28,554 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-03-06 13:28:28,556 INFO mapreduce.Job: Running job: job_local487643619_0001
2026-03-06 13:28:28,581 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-03-06 13:28:28,602 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOut
2026-03-06 13:28:28,605 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-03-06 13:28:28,605 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output
2026-03-06 13:28:28,608 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapreduce.lib.output.FileOutputCo
2026-03-06 13:28:28,751 INFO mapred.LocalJobRunner: Waiting for map tasks
2026-03-06 13:28:28,754 INFO mapred.LocalJobRunner: Starting task: attempt_local487643619_0001_m_000000_0
2026-03-06 13:28:28,811 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOut
2026-03-06 13:28:28,811 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
```

```
2026-03-06 13:28:28,811 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output
2026-03-06 13:28:28,850 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2026-03-06 13:28:28,868 INFO mapred.MapTask: Processing split: file:/exercise2/lines.txt:0+56
2026-03-06 13:28:29,212 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-03-06 13:28:29,212 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-03-06 13:28:29,212 INFO mapred.MapTask: soft limit at 83886080
2026-03-06 13:28:29,213 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-03-06 13:28:29,213 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2026-03-06 13:28:29,220 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-03-06 13:28:29,238 INFO mapred.LocalJobRunner:
2026-03-06 13:28:29,240 INFO mapred.MapTask: Starting flush of map output
2026-03-06 13:28:29,240 INFO mapred.MapTask: Spilling map output
2026-03-06 13:28:29,240 INFO mapred.MapTask: bufstart = 0; bufend = 88; bufvoid = 104857600
2026-03-06 13:28:29,240 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 26214368(104857472); length = 29/65536
2026-03-06 13:28:29,256 INFO mapred.MapTask: Finished spill 0
2026-03-06 13:28:29,274 INFO mapred.Task: Task:attempt_local487643619_0001_m_000000_0 is done. And is in the process of co
2026-03-06 13:28:29,278 INFO mapred.LocalJobRunner: map
2026-03-06 13:28:29,278 INFO mapred.Task: Task 'attempt_local487643619_0001_m_000000_0' done.
2026-03-06 13:28:29,290 INFO mapred.Task: Final Counters for attempt_local487643619_0001_m_000000_0: Counters: 18
	File System Counters
		FILE: Number of bytes read=281830
		FILE: Number of bytes written=992914
		FILE: Number of read operations=0
		FILE: Number of large read operations=0
		FILE: Number of write operations=0
	Map-Reduce Framework
		Map input records=4
		Map output records=8
		Map output bytes=88
		Map output materialized bytes=99
		Input split bytes=90
		Combine input records=8
		Combine output records=7
		Spilled Records=7
		Failed Shuffles=0
		Merged Map outputs=0
		GC time elapsed (ms)=0
		Total committed heap usage (bytes)=432013312
	File Input Format Counters
```

```
!hdfs dfs -cat /exercise2/lines.txt | wc -l
```

```
4
```

EXERCISE 3 : UNIQUE WORD COUNT

```
!echo "spark hadoop spark" > words.txt
!echo "hadoop mapreduce spark" >> words.txt
```

```
!hdfs dfs -mkdir /exercise3
!hdfs dfs -put words.txt /exercise3
```

```
!hadoop jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-*.jar wordcount /exercise3 /output3
```

```
2026-03-06 13:32:03,639 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-03-06 13:32:03,857 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2026-03-06 13:32:03,857 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2026-03-06 13:32:04,192 INFO input.FileInputFormat: Total input files to process : 1
2026-03-06 13:32:04,245 INFO mapreduce.JobSubmitter: number of splits:1
2026-03-06 13:32:04,585 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local117323295_0001
2026-03-06 13:32:04,585 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-03-06 13:32:04,913 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-03-06 13:32:04,914 INFO mapreduce.Job: Running job: job_local117323295_0001
2026-03-06 13:32:04,918 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-03-06 13:32:04,939 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOut
2026-03-06 13:32:04,941 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-03-06 13:32:04,941 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output
2026-03-06 13:32:04,943 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapreduce.lib.output.FileOutputCo
2026-03-06 13:32:05,012 INFO mapred.LocalJobRunner: Waiting for map tasks
2026-03-06 13:32:05,012 INFO mapred.LocalJobRunner: Starting task: attempt_local117323295_0001_m_000000_0
2026-03-06 13:32:05,057 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOut
2026-03-06 13:32:05,058 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-03-06 13:32:05,058 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output
2026-03-06 13:32:05,093 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2026-03-06 13:32:05,109 INFO mapred.MapTask: Processing split: file:/exercise3/words.txt:0+42
2026-03-06 13:32:05,238 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-03-06 13:32:05,238 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-03-06 13:32:05,238 INFO mapred.MapTask: soft limit at 83886080
2026-03-06 13:32:05,238 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-03-06 13:32:05,239 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2026-03-06 13:32:05,270 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-03-06 13:32:05,324 INFO mapred.LocalJobRunner:
2026-03-06 13:32:05,329 INFO mapred.MapTask: Starting flush of map output
2026-03-06 13:32:05,329 INFO mapred.MapTask: Spilling map output
2026-03-06 13:32:05,329 INFO mapred.MapTask: bufstart = 0; bufend = 66; bufvoid = 104857600
```

```
2026-03-06 13:32:05,329 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 26214376(104857504); length = 21/65536
2026-03-06 13:32:05,358 INFO mapred.MapTask: Finished spill 0
2026-03-06 13:32:05,391 INFO mapred.Task: Task:attempt_local117323295_0001_m_000000_0 is done. And is in the process of co
2026-03-06 13:32:05,401 INFO mapred.LocalJobRunner: map
2026-03-06 13:32:05,402 INFO mapred.Task: Task 'attempt_local117323295_0001_m_000000_0' done.
2026-03-06 13:32:05,416 INFO mapred.Task: Final Counters for attempt_local117323295_0001_m_000000_0: Counters: 18
        File System Counters
                FILE: Number of bytes read=281816
                FILE: Number of bytes written=992862
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
        Map-Reduce Framework
                Map input records=2
                Map output records=6
                Map output bytes=66
                Map output materialized bytes=47
                Input split bytes=90
                Combine input records=6
                Combine output records=3
                Spilled Records=3
                Failed Shuffles=0
                Merged Map outputs=0
                GC time elapsed (ms)=43
                Total committed heap usage (bytes)=384827392
        File Input Format Counters
```

```
!hdfs dfs -cat /output3/part-r-00000 | wc -l
```

```
3
```

TOP N FREQUENT WORDS

```
import os

hadoop_home = os.environ["HADOOP_HOME"]
jar_path = hadoop_home + "/share/hadoop/mapreduce/hadoop-mapreduce-examples-*.jar"

# Run WordCount
os.system(f"hadoop jar {jar_path} wordcount /exercise2 /output_top")

# Read output
os.system("hdfs dfs -cat /output_top/part-r-00000 > result.txt")

# Find Top 3 words
word_counts = {}

with open("result.txt") as f:
    for line in f:
        word, count = line.split()
        word_counts[word] = int(count)

top_words = sorted(word_counts.items(), key=lambda x: x[1], reverse=True)[:3]

print("Top 3 Words:")
for w in top_words:
    print(w)
```

```
Top 3 Words:
('data', 2)
('big', 1)
('ecosystem', 1)
```

WORDS APPEARING MORE THAN TWO TIMES

```
import os

# Step 1: Create a new text file
text = """
big data is transforming the world
hadoop is a framework for big data processing
big data analytics helps companies make better decisions
hadoop and mapreduce are important technologies in big data
data science uses statistics machine learning and big data
many companies use hadoop for storing and processing large datasets
big data tools include hadoop spark hive and pig
data engineers work with big data platforms
hadoop distributed file system stores large amounts of data
big data applications are used in healthcare finance and retail
data analysis and big data technologies are growing rapidly
"""
```

```
with open("bigdata_text.txt", "w") as f:
    f.write(text)

print("bigdata_text.txt file created")

# Step 2: Create HDFS directory
os.system("hdfs dfs -mkdir -p /exercise4")

# Step 3: Upload file to HDFS
os.system("hdfs dfs -put -f bigdata_text.txt /exercise4")

print("File uploaded to HDFS")

# Step 4: Run Hadoop WordCount
hadoop_home = os.environ["HADOOP_HOME"]
jar_path = hadoop_home + "/share/hadoop/mapreduce/hadoop-mapreduce-examples-*.jar"

os.system(f"hadoop jar {jar_path} wordcount /exercise4 /output4")

print("Wordcount completed")

# Step 5: Copy MapReduce output to local file
os.system("hdfs dfs -cat /output4/part-r-00000 > word_counts.txt")

# Step 6: Print words appearing more than 2 times
threshold = 2

print("\nWords appearing more than 2 times:\n")

with open("word_counts.txt") as f:
    for line in f:
        word, count = line.split()
        if int(count) > threshold:
            print(word, count)
```

```
bigdata_text.txt file created
File uploaded to HDFS
Wordcount completed

Words appearing more than 2 times:

and 3
big 6
data 8
hadoop 4
```

Problem

Given a text file in HDFS, count how many words start with vowels (a, e, i, o, u).

```
!echo "apple orange banana apple elephant dog india orange apple umbrella eagle banana apple orange dog ice apple elephant'
```

```
!hdfs dfs -mkdir /vowel_ex
```

```
!hdfs dfs -put vowel.txt /vowel_ex
```

```
!hdfs dfs -ls /vowel_ex
```

```
Found 1 items
-rw-r--r--   1 root root        116 2026-03-06 14:25 /vowel_ex/vowel.txt
```

```
!hadoop jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-*.jar wordcount /vowel_ex /vowel_output
```

```
2026-03-06 14:26:19,847 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-03-06 14:26:19,994 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2026-03-06 14:26:19,994 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2026-03-06 14:26:20,388 INFO input.FileInputFormat: Total input files to process : 1
2026-03-06 14:26:20,449 INFO mapreduce.JobSubmitter: number of splits:1
2026-03-06 14:26:20,841 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local742955162_0001
2026-03-06 14:26:20,841 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-03-06 14:26:21,235 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-03-06 14:26:21,237 INFO mapreduce.Job: Running job: job_local742955162_0001
2026-03-06 14:26:21,251 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-03-06 14:26:21,260 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOut
2026-03-06 14:26:21,261 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-03-06 14:26:21,262 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output
2026-03-06 14:26:21,264 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapreduce.lib.output.FileOutputCo
2026-03-06 14:26:21,343 INFO mapred.LocalJobRunner: Waiting for map tasks
2026-03-06 14:26:21,347 INFO mapred.LocalJobRunner: Starting task: attempt_local742955162_0001_m_000000_0
2026-03-06 14:26:21,396 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOut
2026-03-06 14:26:21,396 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
```

```
2026-03-06 14:26:21,398 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output
2026-03-06 14:26:21,429 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2026-03-06 14:26:21,436 INFO mapred.MapTask: Processing split: file:/vowel_ex/vowel.txt:0+116
2026-03-06 14:26:21,774 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-03-06 14:26:21,774 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-03-06 14:26:21,774 INFO mapred.MapTask: soft limit at 83886080
2026-03-06 14:26:21,774 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-03-06 14:26:21,774 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2026-03-06 14:26:21,782 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-03-06 14:26:21,801 INFO mapred.LocalJobRunner:
2026-03-06 14:26:21,803 INFO mapred.MapTask: Starting flush of map output
2026-03-06 14:26:21,803 INFO mapred.MapTask: Spilling map output
2026-03-06 14:26:21,803 INFO mapred.MapTask: bufstart = 0; bufend = 188; bufvoid = 104857600
2026-03-06 14:26:21,803 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 26214328(104857312); length = 69/65536
2026-03-06 14:26:21,821 INFO mapred.MapTask: Finished spill 0
2026-03-06 14:26:21,848 INFO mapred.Task: Task:attempt_local742955162_0001_m_000000_0 is done. And is in the process of co
2026-03-06 14:26:21,856 INFO mapred.LocalJobRunner: map
2026-03-06 14:26:21,857 INFO mapred.Task: Task 'attempt_local742955162_0001_m_000000_0' done.
2026-03-06 14:26:21,867 INFO mapred.Task: Final Counters for attempt_local742955162_0001_m_000000_0: Counters: 18
        File System Counters
                FILE: Number of bytes read=281889
                FILE: Number of bytes written=992940
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
        Map-Reduce Framework
                Map input records=1
                Map output records=18
                Map output bytes=188
                Map output materialized bytes=118
                Input split bytes=89
                Combine input records=18
                Combine output records=9
                Spilled Records=9
                Failed Shuffles=0
                Merged Map outputs=0
                GC time elapsed (ms)=0
                Total committed heap usage (bytes)=371195904
        File Input Format Counters
```

```
!hdfs dfs -cat /vowel_output/part-r-00000
```

```
apple   5
banana  2
dog     2
eagle   1
elephant        2
ice     1
india   1
orange  3
umbrella        1
```

```
!hdfs dfs -cat /vowel_output/part-r-00000 | grep -E "^[aeiouAEIOU]"
```

```
apple   5
eagle   1
elephant        2
ice     1
india   1
orange  3
umbrella        1
```

Problem

Count how many words in the file have length greater than 5.

```
import os

with open("length_input.txt","w") as f:
    f.write("hadoop spark distributed computing python bigdata machinelearning")

os.system("hdfs dfs -mkdir -p /length_example")
os.system("hdfs dfs -put -f length_input.txt /length_example")

# mapper
with open("mapper.py","w") as f:
    f.write("""
import sys
for line in sys.stdin:
    words=line.strip().split()
    for w in words:
        if len(w)>5:
            print("longword\\t1")
""")
```

```
# reducer
with open("reducer.py","w") as f:
    f.write("""
import sys
count=0
for line in sys.stdin:
    key,val=line.strip().split("\\t")
    count+=int(val)
print("Words longer than 5:",count)
""")

os.system("hadoop jar /usr/lib/hadoop-mapreduce/hadoop-streaming.jar \
-input /length_example/length_input.txt \
-output /length_output \
-mapper mapper.py \
-reducer reducer.py")

os.system("hdfs dfs -cat /length_output/part-00000")
```

```
256
```

Problem

Count how many words start with a, e, i, o, u individually.

```
import os

with open("vowelcount_input.txt","w") as f:
    f.write("apple ant elephant eagle ice orange umbrella owl")

os.system("hdfs dfs -mkdir -p /vowelcount_example")
os.system("hdfs dfs -put -f vowelcount_input.txt /vowelcount_example")

# mapper
with open("mapper.py","w") as f:
    f.write("""
import sys
vowels=('a','e','i','o','u')
for line in sys.stdin:
    words=line.strip().lower().split()
    for w in words:
        if w.startswith(vowels):
            print(w[0]+"\\t1")
""")

# reducer
with open("reducer.py","w") as f:
    f.write("""
import sys
current=None
count=0

for line in sys.stdin:
    key,val=line.strip().split("\\t")
    val=int(val)

    if key==current:
        count+=val
    else:
        if current:
            print(current,count)
        current=key
        count=val

if current:
    print(current,count)
""")

os.system("hadoop jar /usr/lib/hadoop-mapreduce/hadoop-streaming.jar \
-input /vowelcount_example/vowelcount_input.txt \
-output /vowelcount_output \
-mapper mapper.py \
-reducer reducer.py")

os.system("hdfs dfs -cat /vowelcount_output/part-00000")
```

```
256
```

STOCK ANALYSIS PROBLEM

```
from google.colab import files
files.upload()
```

Choose files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving goog.csv to goog (1).csv

{'goog (1).csv': b'Date,Close,High,Low,Open,Volume\n2004-08-
19,2.4805288314819336,2.572490948053174,2.3722498936767993,2.472122773210547,897427216\n2004-08-
20,2.6775567531585693,2.6965918992497553,2.4844838896751833,2.4970913195361195,458857488\n2004-08-
23,2.704503059387207,2.8053655894795213,2.695850891086642,2.7378766889277246,366857939\n2004-08-
24,2.592515468597412,2.7588887660973636,2.5603775980873573,2.7499895432413908,306396159\n2004-08-
25,2.620450258255005,2.6698934124941425,2.5680412483570305,2.5947401020231093,184645512\n2004-08-
26,2.667668342590332,2.668657040145922,2.5873240057583904,2.5944931279332617,142572401\n2004-08-
27,2.6241583824157715,2.685220267763395,2.6127867057755947,2.6723648350876354,124826132\n2004-08-
30,2.521812915802002,2.6078428451983524,2.521812915802002,2.6026506454500833,104429967\n2004-08-
31,2.530712604522705,2.5638389412984854,2.525521350880501,2.528982502166952,98825037\n2004-09-
01,2.478303194046020 5,2.5455447910407454,2.463964715710837,2.538870018881148,183633734\n2004-09-
02,2.5094518661499023,2.530712411077297,2.445918233984355,2.4520988946153928,303810504\n2004-09-
03,2.472369432449341,2.5151378087091354,2.455312392392517,2.495608315447998,103538639\n2004-09-
07,2.5111825466156006,2.521565762493027,2.462481981902271,2.4970913605126936,117506800\n2004-09-
08,2.528981924057007,2.5470286079021105,2.484483682707569,2.490416103175605,100186120\n2004-09-
09,2.529229164123535,2.539117084458438,2.4968442372196527,2.5346674729789918,81620792\n2004-09-
10,2.6038870811462402,2.634294138915772,2.5042603239383383,2.511676737209856,174804764\n2004-09-
13,2.657532215118408,2.680028514126486,2.6318220576904374,2.6360246135905094,157628624\n2004-09-
14,2.7561705112457275,2.7687779422366785,2.6399798630744873,2.6562964477810356,217608605\n2004-09-
15,2.768777847290039,2.823906367128911,2.7242795997712306,2.7331797698911653,215279909\n2004-09-
16,2.8174781799316406,2.862717828452675,2.7601254505784443,2.7771824914931575,186207345\n2004-09-
17,2.9044976234436035,2.9044976234436035,2.8070953112741055,2.8286029110103463,190350817\n2004-09-
20,2.950726270675659,3.0061018497481564,2.886698523597279,2.891148135626561 4,213585582\n2004-09-
21,2.913149833679199,2.976930519174107,2.904992015129819,2.961850635519053,145262446\n2004-09-
22,2.9264988899230957,2.9583897083696775,2.887686952422269,2.902272725117842,152344894\n2004-09-
23,2.9868195056915283,3.031564809736841,2.892878102561242,2.9378716458662635,171524515\n2004-09-
24,2.9623448848724365,3.067904066917511,2.9606147827805325,2.9897849051250365,183336625\n2004-09-
27,2.9235332012176514,2.9883023508389788,2.91216152387903,2.955670128374949,141994242\n2004-09-
28,3.1361353397369385,3.14948417435316,2.971739432874207,2.998685341350608,340190355\n2004-09-
29,3.2404589653015137,3.3378603288205086,3.1205610684841716,3.1321800374887006,613229953\n2004-09-
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```
os.system("hdfs dfs -mkdir -p /finance_data")
```

0
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```
os.system("hdfs dfs -put -f goog.csv /finance_data")
```

12,4.499264240264893,4.692090073683194,4.385546987170407,4.579114462050222,672354740\n2004-11-
0
15,4.570213794708252,4.655502080228841,4.418920372858908,4.460945928966434,477844390\n2004-11-
16,4.2654008865356445,4.436719110185026,4.223128271626627,4.388018305170045,839832142\n2004-11-

```
os.system("hdfs dfs -ls /finance_data")
```

19,4.187775611877441,4.202114326819459,4.116578515645506,4.180359198385845,352086780\n2004-11-
0
22,4.081475257873535,4.190248546086964,3.9877823276818383,4.065901261041375,496582362\n2004-11-
23,4.141300678253174,4.223128291943862,4.116084869527423,4.152425299749828 5,498393124\n2004-11-

```
os.system("hdfs dfs -cat /finance_data/goog.csv > stock_local.csv")
```

29,4.475779056549072,4.522749049439898,4.388265367733138,4.458721303644271,428263241\n2004-11-
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01,4.448833465576172,4.511624512446362,4.438697541405131,4.49802838367995,315743063\n2004-12-

```
import pandas as pd

df = pd.read_csv("stock_local.csv")

print("Dataset Preview")
print(df.head())

print("\nAverage Closing Price:", df["Close"].mean())
print("Maximum Closing Price:", df["Close"].max())
```

```
print("Minimum Closing Price:", df["Close"].min())

print("\nTotal Volume:", df["Volume"].sum())
```

```
Dataset Preview
         Date     Close      High       Low      Open     Volume
0  2004-08-19  2.480529  2.572491  2.372250  2.472123  897427216
1  2004-08-20  2.677557  2.696592  2.484484  2.497091  458857488
2  2004-08-23  2.704503  2.805366  2.695851  2.737877  366857939
3  2004-08-24  2.592515  2.758889  2.560378  2.749990  306396159
4  2004-08-25  2.620450  2.669893  2.568041  2.594740  184645512

Average Closing Price: 56.09455822214581
Maximum Closing Price: 336.42999267578125
Minimum Closing Price: 2.47236943244934l

Total Volume: 589327768393
```

22,4.605566024780273,4.619162153856454,4.524232993368904,4.54623470786622,156865776\n2004-12-
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27,4.744251251220703,4.778613812280789,4.674784487687093,4.676021187759403,245079187\n2004-12-
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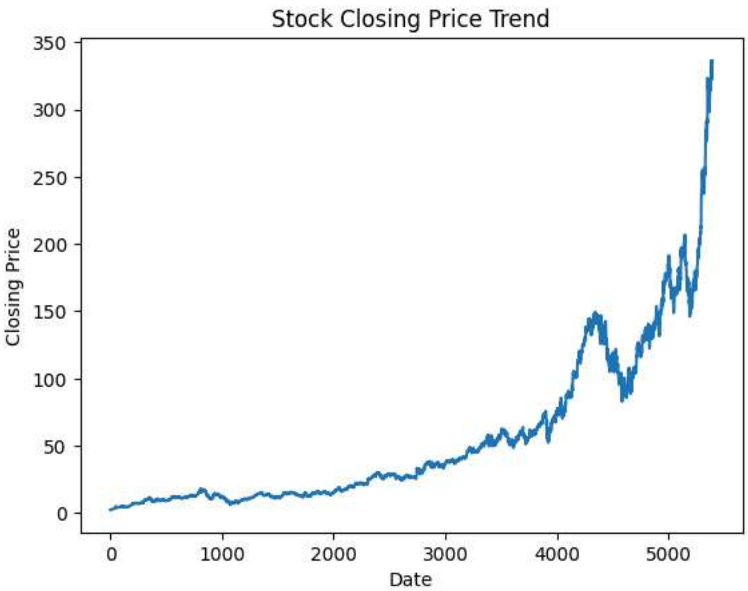
```
import matplotlib.pyplot as plt

plt.figure()

plt.plot(df["Close"])

plt.title("Stock Closing Price Trend")
plt.xlabel("Date")
plt.ylabel("Closing Price")

plt.show()
```



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```
plt.figure()
plt.bar(range(len(df)), df["Volume"])
plt.title("Daily Trading Volume")
plt.xlabel("Date")
plt.ylabel("Volume")
plt.show()
```

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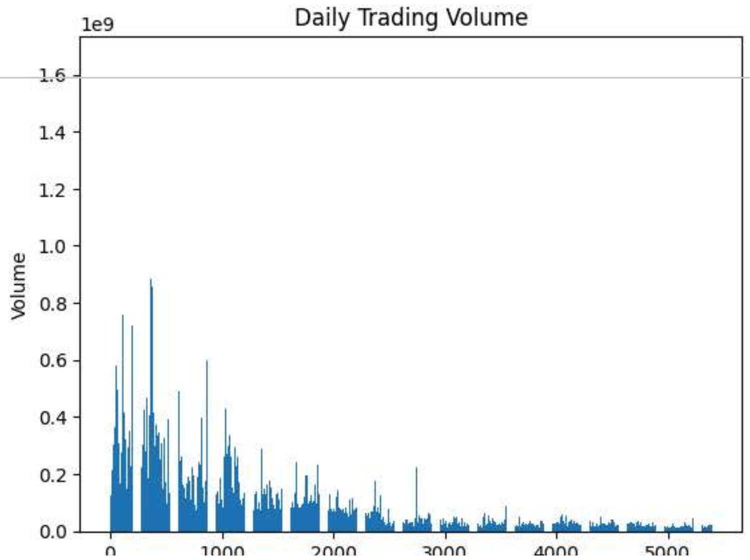


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